

Software Engineering

Chapter 3- (Project Management Concepts)

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The Management Spectrum

Effective software project management focuses on the four P's:

People — the most important element of a successful project

Product — the software to be built

Process — the set of framework activities and software engineering tasks to get the job done

Project — all work required to make the product a reality

The People

- People of project include from manager to developer, from customer to end user.
- But mainly people of a project highlight the developers.
- It is so important to have highly skilled and motivated developers that the Software Engineering Institute (SEI) has developed a ***people management capability maturity model (PM-CMM)***.
- Organization that achieve high levels of maturity in the people management area have a higher likelihood of implementing effective software engineering practices.

The People

The people management maturity model defines: recruiting, selection, performance management, training, compensation, career development, organization and work design, and team/culture development.

The Product

- The product is the ultimate goal of the project. This is any type of software product that has to be developed.
- To develop a software product successfully, all the product objectives and scopes should be established, alternative solutions should be considered, and technical and management constraints should be identified beforehand.
- Lack of these information, it is impossible to define reasonable and accurate estimation of the cost, an effective assessment of risks, a realistic breakdown of project tasks or a manageable project schedule that provides a meaningful indication of progress.

The Process

A software process provides the framework for which a comprehensive plan for software development can be established.

Task sets – tasks, milestones, work products, and quality assurance points

Umbrella activities – software quality assurance, software configuration management, and measurement

The Project

- The project is the complete software project that include the requirement analysis, development, delivery, maintenance and updates.
- The project manager of a project or sub-project is responsible for managing the people, product and process. The responsibilities or activities of software project manager would be a long list but that has to be followed to avoid project failure.

The Project

To manage complexity

To avoid failure

To develop a common sense approach for planning, monitoring, and controlling the project.

People

The Players

Team Leaders

The Software Team

Coordination and Communication Issues

The Players

Five categories:

1. Senior Managers: defines business issues
2. Project (technical) managers: plan, motivate, organize, and control the practitioners
3. Practitioners: deliver the technical skills
4. Customers: specify the requirements for the software
5. End-Users: interact with the software once it is released

Team Leaders

Project management is a people-intensive activity → need “people skill”

MOI model for Leadership:

- Motivation
- Organization
- Ideas for innovation

The Software Team

N individuals vs. m tasks

Team organizations

- Democratic decentralized (DD): no permanent leader, rather “task coordinator”, decision made by group consensus.
- Controlled decentralized (CD): has defined leader, decision remains group activity, works partitioned
- Controlled centralized (CC): Top-level problem solving, internal coordination

The Software Team

Seven project factors when planning the structure of software engineering team:

The difficulty of the problem

The size of the resultant program

The time

The degree of problem to be modularized

The required quality and reliability

The rigidity of the delivery date

Degree of sociability (communication)

Coordination and Communication Issues

Many reasons that software projects get into trouble:

Scale

Uncertainty

Interoperability

Therefore, must establish methods for coordinating the people.

Coordination and Communication Issues

Hence, establish formal and informal communication among team members:

Formal, impersonal approaches: SE docs and deliverables, tech memo.

Formal, interpersonal procedures: QA activities, status review meetings and design

Informal, interpersonal procedures: group meeting

Electronic communication: email, e-bbs

Interpersonal networking: interpersonal discussion with outsiders.

The Product

Dilemma: Quantitative estimates, but no solid information

1. Software scope
2. Problem decomposition

Software Scope

Context:

Information objectives

Function and performance

SW scope must be unambiguous and understandable.

Problem Decomposition

Sometimes call partitioning or problem elaboration

The core of software requirement analysis

1. Functionality
2. Process

The Process

The generic phases that characterize the software process – definition, development, and support – are applicable to all software.

The problem is to select the process that is appropriate for the software to be engineered by a project team.

The Process

The linear sequential model

The prototyping model

The RAD model

The incremental model

The spiral model

The component-based development model

The concurrent development model

The formal methods

The fourth generation techniques model

The Process

Must decide which model is most appropriate for

1. the customers
2. The characteristics of the product
3. The project environemnt

The Project

Must understand what can go wrong (so that problems can be avoided)

Ten signs that indicate that an information systems project is in jeopardy:

1. Software people don't understand their customer's needs
2. The product scope is poorly defined
3. Changes are managed poorly

Cont...

The Project

Ten signs (cont..)

4. The chosen technology changes
5. Business needs change (or ill-defined)
6. Deadlines are unrealistic
7. Users are resistant
8. Sponsorship is lost (or was never properly obtained)
9. The project team lacks people with appropriate skills
10. Managers (and practitioners) avoid best practices and lessons learned

The Project

Five-part commonsense approach to software project:

1. **Start on the right foot:** working hard to understand the problem
2. **Maintain momentum:** provide incentives
3. **Track progress:** track work products
4. **Make smart decisions:** decisions should be “keep it simple”
5. **Conduct a postmortem analysis:** lessons learned and evaluation of project

The W⁵HH Principle

Barry Boehm suggests an approach that addresses project objectives, milestones and schedules, responsibilities, management and technical approaches, and requires resources:

- Why is the system being developed?
- What will be done, by when?
- Who is responsible for a function?
- Where are they organizationally located?
- How will the job be done technically and managerially?
- How much of each resources is needed?

Thank You